Server Pulse setup

Copy and paste each block into your server's terminal, in order. At the end you get one link to send back. It lets the Server Pulse dashboard show this server's CPU, RAM, disk, network and temperature live, from anywhere.

Ubuntu 22.04 or 24.04 | A user with **sudo** | About **10 minutes** | No router ports opened

First, pick how the dashboard will reach this server

Tailscale Funnel
Recommended. Free, fixed address, no request limit. Needs a free Tailscale account.

ngrok
Free account with one free domain. Live view is limited to about 54 hours a month.

The steps below change to match your choice. If you're unsure, use Tailscale.

1 Create your two passwords

This makes two random passwords and saves them in a file only you can read. Every later step reads them from there, so nothing needs editing. **owner** is for you in the browser; **dashboard** is only for the Server Pulse app.

Copy

```
[ -f ~/.server-pulse ] || {
  printf "OWNER_PASS=%s\nAPP_PASS=%s\n" \
    "$(tr -dc A-Za-z0-9 </dev/urandom | head -c 20)" \
    "$(tr -dc A-Za-z0-9 </dev/urandom | head -c 20)" > ~/.server-pulse
  chmod 600 ~/.server-pulse
}
cat ~/.server-pulse
```

You should see two lines: `OWNER_PASS=…` and `APP_PASS=…`.

2 Install btop and show it in the browser

btop is a live view of everything running on the server. ttyd shows it as a read-only web page, only on this machine for now.

```
sudo apt update && sudo apt install -y btop curl wget python3
sudo wget -qO /usr/local/bin/ttyd "https://github.com/tsl0922/ttyd/releases/downloa
sudo chmod +x /usr/local/bin/ttyd
```

```
sudo tee /etc/systemd/system/btop-web.service > /dev/null <<EOF
[Unit]
Description=btop live in browser (ttyd)
After=network-online.target
Wants=network-online.target

[Service]
User=$USER
Environment=TERM=xterm-256color
ExecStart=/usr/local/bin/ttyd -p 7681 -i lo -t titleFixed=btop-live -t fontSize=14
Restart=always
RestartSec=3

[Install]
WantedBy=multi-user.target
EOF

sudo systemctl daemon-reload && sudo systemctl enable --now btop-web
systemctl is-active btop-web
```

Last line should say `active`.

3 Install Netdata (records every second, keeps 30 days)

Netdata measures the server every second and keeps 30 days on disk. The dashboard reads from it.

```
wget -qO /tmp/netdata-kickstart.sh https://get.netdata.cloud/kickstart.sh
sh /tmp/netdata-kickstart.sh --stable-channel --disable-telemetry --non-interactive

sudo tee -a /etc/netdata/netdata.conf > /dev/null <<'EOF'

[web]
    bind to = 127.0.0.1

[db]
    mode = dbengine
    dbengine tier 0 retention time = 30d
EOF

sudo systemctl restart netdata
sleep 5
curl -s -o /dev/null -w '%{http_code}\n' http://127.0.0.1:19999/api/v1/info
```

Last line should print `200`.

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4 Put a login in front of both

Caddy asks for a password before showing anything, and compresses data so the dashboard stays fast.

```
source ~/.server-pulse
sudo apt install -y debian-keyring debian-archive-keyring apt-transport-https gnupg
curl -1sLf 'https://dl.cloudsmith.io/public/caddy/stable/gpg.key' | sudo gpg --dea
curl -1sLf 'https://dl.cloudsmith.io/public/caddy/stable/debian.deb.txt' | sudo te
sudo apt update && sudo apt install -y caddy

OWNER_HASH=$(caddy hash-password --plaintext "$OWNER_PASS")
APP_HASH=$(caddy hash-password --plaintext "$APP_PASS")

sudo tee /etc/caddy/Caddyfile > /dev/null <<EOF
:8080 {
    bind 127.0.0.1
    encode zstd gzip
    basic_auth {
        owner $OWNER_HASH
        dashboard $APP_HASH
    }
    redir /history /history/
    handle_path /history/* {
        reverse_proxy 127.0.0.1:19999
    }
    handle {
        reverse_proxy 127.0.0.1:7681
    }
}
EOF

caddy validate --config /etc/caddy/Caddyfile --adapter caddyfile
sudo systemctl enable --now caddy && sudo systemctl restart caddy
sleep 2
curl -s -o /dev/null -w '%{http_code}\n' -u "dashboard:$APP_PASS" http://127.0.0.1
```

Should end with `Valid configuration` and then `200`.

5 Open a private HTTPS link with Tailscale

Run these one at a time. Each of the first two prints a link: open it in your browser, log in or approve, then come back. If `funnel` asks you to enable HTTPS and Funnel, open its link, approve, and run the same command again.

```
curl -fsSL https://tailscale.com/install.sh | sh
sudo tailscale up


sudo tailscale funnel --bg 8080
```

Needed for 24/7: open login.tailscale.com/admin/machines, click ⋯ next to this server, choose 180 days.

```
tailscale funnel status
tailscale status --json | python3 -c 'import json,sys; print(json.load(sys.stdin)['
```

You should see an `https://….ts.net` address and `key expiry disabled ✓`.

6 Keep the machine awake 24/7

A server that sleeps drops its link. This turns off sleep and stops a closed laptop lid from suspending it. It's harmless on a cloud server.

```
sudo systemctl mask sleep.target suspend.target hibernate.target hybrid-sleep.targ
sudo mkdir -p /etc/systemd/logind.conf.d
sudo tee /etc/systemd/logind.conf.d/server-pulse.conf > /dev/null <<'EOF'
[Login]
HandleLidSwitch=ignore
HandleLidSwitchExternalPower=ignore
HandleLidSwitchDocked=ignore
EOF
systemctl is-enabled sleep.target suspend.target
```

Should print `masked` twice.

7 Get your link

This prints the finished link and checks it works from the internet.

```
source ~/.server-pulse
if [ -n "$NGROK_DOMAIN" ]; then HOST="$NGROK_DOMAIN"
else HOST=$(tailscale status --json | python3 -c 'import json,sys; print(json.load
echo
echo "Link works? $(curl -s -o /dev/null -w '%{http_code}' -u "dashboard:$APP_PASS'
echo
echo "SEND THIS LINK:  https://dashboard:$APP_PASS@$HOST/history"
echo
```

```
echo "btop in your browser: https://$HOST   user: owner   password: $OWNER_PASS"
```

Look for `Link works? 200`. A brand-new Tailscale address can take a minute to come online; run the

8 Check it survives a reboot

Everything starts by itself at boot. Reboot once, log back in, and check.

```
sudo reboot
```
Copy

```
systemctl is-active btop-web netdata caddy
```
Copy

Should print `active` three times. Then run step 7 again: you should still get `200`.

Done: send the link

Send the `https://dashboard:…/history` link from step 7 back privately (direct message, not a public group). It contains a read-only password: it can show the server's stats but can't change anything on the server.

Whoever runs the dashboard pastes it in **Server Pulse** → **Connect** (or as `SERVER_URL` in Vercel), and live data appears within seconds.

If something doesn't work

Step 4 shows `401`	The passwords don't match. Run step 4 again.
Step 4 shows `502`	Netdata isn't running: `sudo systemctl restart netdata`, wait 10 seconds, run step 4's last line again.
Step 7 shows `000`	The tunnel isn't up yet. Tailscale: `tailscale funnel status`. ngrok: `systemctl status ngrok-pulse`, and check the token and domain.
Lost the passwords	`cat ~/.server-pulse` shows them again.
Stops working after months (Tailscale)	The key expired. Run `sudo tailscale up`, then disable key expiry as in step 5.

Stops when the laptop sleeps	Run step 6 and reboot.